PLN and **NARS**

PLN and **NARS** — that both try to handle a hard problem: *how should a machine reason logically when it isn't sure about its facts?* even though the two systems are built on completely different philosophical foundations, they often **arrive at nearly the same answers** in realistic conditions just by very different routes.

The problem

AI needs to reason under uncertainty. Real-world reasoning is rarely clean you usually have partial information and have to make judgments anyway this is essential for AGI.

Q. What do we mean by the uncertainty

Uncertainty means *not knowing something for sure*. It's the gap between "I know this is true" and "I have no idea." Almost everything we reason about in real life sits somewhere in that gap.

Example: "It will rain tomorrow" → you're not certain, just guessing based on clouds

Existing tools fall short in different ways:

- **Bayes Nets and Markov Logic Networks** are good at probability math but weak at expressive logic. They struggle with statements like "for all X" (quantifiers) or statements *about* statements (higher-order expressions).
- **LLMs** (like ChatGPT) are impressive but make obvious logical errors despite their fluency.

The two systems being compared:

PLN (Probabilistic Logic Networks) — Ben Goertzel's framework. It blends:

- Probabilistic reasoning
- Predicate logic (with quantifiers like "for all" and "there exists")
- Term logic (relationships between categories — e.g. "all swans are birds")
- An *observation-grounded* meaning: truth values come from what's actually been observed, not just abstract assumption

NARS (Non-Axiomatic Reasoning System) — Pei Wang's framework. Key features:

- It's a *term logic* — reasoning about relationships between concepts
- It tracks uncertainty through **evidence**, not probability
- "Non-axiomatic" means it doesn't assume any fixed truths it works with limited knowledge, limited resources, the way humans actually do

Both support the three classical inference patterns:

- **Deduction** — General to specific. "All men are mortal. Socrates is a man. So Socrates is mortal." **Definition:** *If the premises are true, then the conclusion is guaranteed to be true.*

- **Induction** — Specific to general. "Every swan I've seen is white, so probably all swans are white." **Definition:** *If the premises are true, then the conclusion is probably true — but not guaranteed.*

- **Abduction** — Effect to likely cause. "The grass is wet. It probably rained." **Definition:** *If the premises are true, then the conclusion is the best available explanation — but not guaranteed.*

Type	**Direction**	**Certainty**	**Example Logic**
Deduction	General → Specific	Guaranteed if premises are true	Rule + Case → Result
Induction	Specific → General	Probable	Cases + Results → Rule
Abduction	Effect → Cause	Best guess	Rule + Result → Case

Both also extend these basics to handle harder scenarios: quantifiers, causation, time, etc. And critically, both attach **two numbers** to every statement:

- **Strength** — how true it seems
- **Confidence** — how much evidence backs that judgment

Fundamental Difference: PLN vs NARS

Point	PLN	NARS
Full form	Probabilistic Logic Networks	Non-Axiomatic Reasoning System
Where uncertainty exists	On terms/concepts and relationships	Only on statements/relationships
Can a concept alone have probability?	Yes	No
Example: probability of bird	PLN can say: "random object has 5% chance of being a bird"	NARS does not ask this directly
Main question asked	"How probable is this concept or relation?"	"How much evidence supports this statement?"
Meaning of uncertainty	Probability-style uncertainty	Evidence-based uncertainty
Example relationship	Bird → Fly has some probability/truth value	<bird --> fly> has frequency and confidence
Philosophical style	More like probabilistic logic / statistical reasoning	More like experience-grounded reasoning under limited knowledge

Rather than compare strength and confidence separately across the two systems, they multiply them into a single number representing the *overall weight* you should give to a conclusion.

Power = Strength × Confidence

When term probabilities are highly uncertain (i.e., PLN doesn't really know how common terms are in the world), the two systems produce nearly the same **power** for the conclusions of inferences even though they computed it through completely different math.

When PLN *does* have good information about term probabilities (low uncertainty about terms), the agreement collapses especially for induction and abduction. In that regime, PLN becomes **strongly Bayesian** it leans heavily on prior probabilities the way classical Bayes' theorem does. NARS, having no term probabilities to lean on, can't follow PLN there. So they diverge.

Q. Why these matters

The result suggests that **two rival philosophical approaches to uncertain reasoning may secretly be capturing the same underlying truth** at least in the messy, realistic case where we don't know much about how common things are in the world. It's a kind of convergence result: when you're genuinely ignorant about the world (which is most of the time), it doesn't really matter whether you reason like a Bayesian probabilist or like an evidence-counter you'll come to roughly the same overall confidence in your conclusions, even if you describe that confidence with different numbers.

Non-Axiomatic Reasoning System (NARS)

The Non-Axiomatic Reasoning System (NARS) is a reasoning framework developed by Pei Wang, designed to enable artificial intelligence systems to reason intelligently under uncertainty. Unlike classical logic, which assumes complete and certain knowledge, and unlike standard probabilistic systems, which assume known probability distributions, NARS is built on a more realistic premise: an intelligent system must operate with **insufficient knowledge and limited resources**. This philosophical foundation sometimes called the *Assumption of Insufficient Knowledge and Resources (AIKR)* shapes every aspect of how NARS represents and processes information.

In simple:

NARS is a term logic system, meaning it reasons about relationships between concepts (terms) rather than between full propositions. Statements in NARS take forms like "ravens are birds" or "birds can fly," where the system tracks how strongly evidence supports each such relationship. Crucially, **NARS** does not assume any prior probabilities about the terms themselves; it only tracks uncertainty in the relationships between them, based on accumulated experience.

Representing Uncertainty:

Every statement in NARS is attached to a truth value represented as a pair **(f, c)**:

Frequency (f): The proportion of positive evidence supporting the statement. For example, if 90 out of 100 observed ravens were black, then f = 0.9 for "ravens are black."

Confidence (c): A measure of how much evidence the system has, calculated as $c = n / (n + k)$, where n is the number of evidential observations and k is a system parameter. Confidence starts at 0 with no evidence and approaches 1 as evidence accumulates, but never reaches absolute certainty.

How NARS attaches numbers to uncertainty:

Every statement in NARS gets two values attached: (f, c) frequency and confidence. These two numbers capture different aspects of uncertainty

Where:

Frequency (f) — "How often is this true?"
f = proportion of positive evidence supporting a statement

This is just a ratio. Out of all the evidence you've collected about a statement, what fraction supports it being true?

Example: Suppose you're tracking the statement "ravens are black."

- You've observed 100 ravens total.
- 90 of them were black, 10 were some other colour.
- Then f = 90/100 = **0.9**

So f = 0.9 means "90% of my evidence supports this statement." It's a clean, intuitive number.

Confidence (c) — "How much evidence do I have?"
c = n / (n + k)

Where:

n = number of evidential observations supporting the statement
k = an *experience parameter* (a tuning constant chosen by the system designer, often k = 1 or k = 2)

what it does with k = 1:

n (observations)	c (confidence)
0	0/1 = 0.00
1	1/2 = 0.50
5	5/6 = 0.83
10	10/11 = 0.91
100	100/101 = 0.99
1000	1000/1001 = 0.999
∞	approaches 1.00

Notice three important properties:

1. **Confidence starts at 0** with no evidence you have no right to be confident in anything you haven't observed.

2. **Confidence grows toward 1** as evidence accumulates more evidence makes you more sure.

3. **Confidence never reaches 1** you can never be *perfectly* certain. This is a deliberate design choice that reflects the philosophy of NARS: an intelligent system should always remain open to new evidence overturning its beliefs.

The parameter **k** controls how quickly confidence grows. A small k (like 1) makes the system gain confidence fast. A larger k makes it more cautious it requires more evidence before being confident.

The "Expectation" concept:

Expectation = f × c

For example:

f = 0.9, c = 0.5 → Expectation = 0.45 (looks true, but only moderately supported)
f = 0.9, c = 0.99 → Expectation = 0.89 (looks true and very well-supported)
f = 0.5, c = 0.99 → Expectation = 0.495 (genuinely uncertain, but you've checked carefully)

In probability theory, "expectation" has a very specific meaning — *expected value*, the average outcome of a random variable

NARS's "Expectation = $f \times c$" has **nothing to do with** that probabilistic concept. It's just a convenience product. But since the paper is comparing NARS to PLN (which *does* use "expectation" in the standard probabilistic sense), using the same word for two different things would be a recipe for confusion.

Solution: rename it to "Power"
Power = $f \times c$

Deduction:

In term logic, deduction is transitive inference chaining inheritance relationships together.

If $A \rightarrow B$ (A inherits from B, i.e. "A is a kind of B")
And $B \rightarrow C$ (B inherits from C, i.e. "B is a kind of C")
Then $A \rightarrow C$ (A is a kind of C)

Example:

$A \rightarrow B$: "Ravens are birds" (with some uncertainty)
$B \rightarrow C$: "Birds are animals" (with some uncertainty)
$A \rightarrow C$: "Ravens are animals" (the conclusion, with derived uncertainty)

Each premise comes with its own (frequency, confidence) pair:
$A \rightarrow B$ has truth value **(f_1, c_1)**
$B \rightarrow C$ has truth value **(f_2, c_2)**

The job of the deduction formula is to compute the conclusion's truth value:
$A \rightarrow C$ has truth value **(f, c)**

Frequency of the conclusion

$$f = \frac{f_1 f_2}{f_1 + f_2 - f_1 f_2}$$

A quick check at the extremes:

If $f_1 = 1$ and $f_2 = 1 \rightarrow f = 1/(1+1-1) = 1$
(both certain → conclusion certain in frequency)

If $f_1 = 1$ and $f_2 = 0 \rightarrow f = 0/(1+0-0) = 0$
(one premise totally false → conclusion has no support)

If $f_1 = 0.5$ and $f_2 = 0.5 \rightarrow f = 0.25/0.75 \approx 0.33$
(some support, but weakened)

The formula essentially captures "how much positive evidence flows through the chain," accounting for the way NARS bookkeeps evidence rather than probabilities.

Confidence of the conclusion

$$c = c_1 c_2 (f_1 + f_2 - f_1 f_2)$$

Two things to notice:

1. **It contains c_1 × c_2** confidence multiplies, so the conclusion can never be more confident than either premise. This makes sense: a chain of reasoning is only as trustworthy as its weakest link.
2. **It's also multiplied by ($f_1 + f_2 - f_1 f_2$)** this factor adjusts the confidence based on how strongly the frequencies support the chain. If both frequencies are high, this factor is close to 1 (you keep most of your confidence). If the frequencies are low, the factor is small, so confidence drops.

Quick check:

- If $c_1 = c_2 = 1$ and $f_1 = f_2 = 1 \rightarrow c = 1 \times (1+1-1) = 1$ (max confidence)
- If $c_1 = 0.5$, $c_2 = 0.5$, $f_1 = f_2 = 1 \rightarrow c = 0.25 \times 1 = 0.25$ (confidence drops because of weak premises)
- Any $c = 0$ in premises → $c = 0$ in conclusion (no evidence in → no evidence out)

So the chain *erodes* confidence that's deduction working with uncertainty. The more steps in your chain, the less confident the final conclusion (which matches intuition: long chains of inference get shakier).

Power of the conclusion

$$\text{Power} = f \times c = f_1 f_2 c_1 c_2$$

This is the elegant punchline. When you multiply the frequency and confidence formulas together, the messy denominator and the ($f_1 + f_2 - f_1 f_2$) factor cancel out, and you're left with just:

This is *really* clean. It says:
The overall reliability of the conclusion is just the product of the reliabilities of the premises.
It matches the intuition perfectly: when you chain two beliefs together, you multiply their reliabilities. If each premise has Power = 0.8, the conclusion has Power = 0.64. Each link in the chain compounds the

uncertainty multiplicatively.
This is also why the authors care about Power so much. It's the *single number* that comes out cleanly from NARS's messier two-number formulas and it's the number they'll compare against PLN.

Example

Premises:

- "Ravens are birds": $(f_1, c_1) = (0.95, 0.9)$ — strong evidence, mostly true
- "Birds are animals": $(f_2, c_2) = (0.99, 0.95)$ — very strong, almost certain

Conclusion ("Ravens are animals"):

- $f = (0.95 \times 0.99) / (0.95 + 0.99 - 0.95 \times 0.99)$
  - Numerator: 0.9405
  - Denominator: $0.95 + 0.99 - 0.9405 = 0.9995$
  - $f \approx$ **0.941**
- $c = 0.9 \times 0.95 \times 0.9995 =$ **0.855**
- Power $= 0.941 \times 0.855 \approx$ **0.804**
  - check: $f_1 f_2 c_1 c_2 = 0.95 \times 0.99 \times 0.9 \times 0.95 = 0.804$

So "Ravens are animals" comes out with frequency ≈ 0.941 and confidence ≈ 0.855 — slightly less than either premise, as expected

Induction:

Induction is a type of reasoning where we move from **specific examples** to a **general rule**.

Induction means learning a general pattern from repeated examples or observations.

The setup

- Premise 1: **B → A** with (f_1, c_1) — "Bs are As" (with its evidence)
- Premise 2: **B** with (f_2, c_2) — evidence about the term B itself

Conclusion:

- **A** with (f, c) — what we can say about A

Frequency

$$f = f_1$$

The conclusion's frequency is just inherited directly from the first premise's frequency.

Confidence

$$c = \frac{f_2 c_2 c_1}{f_2 c_2 c_1 + k}$$

- $\mathbf{c_1}$ — how confident we are in the B→A relationship
- $\mathbf{c_2}$ — how confident we are in B
- $\mathbf{f_2}$ — how true B itself looks (frequency of B)

Multiplying these together gives the "amount of effective evidence" supporting the inductive leap. Then plugging that into the standard *n/(n+k)* shape gives the confidence.

Power

$$\text{Power}_{\text{induction}} = \frac{f_1 \cdot (f_2 c_1 c_2)}{(f_2 c_1 c_2) + k}$$

Notice the structure: it's f_1 **multiplied by another n/(n+k)-shaped thing**.

Example:

Premises:

- B → A: "Ravens are black" with $(f_1, c_1) = (0.9, 0.8)$
- B: "Raven" observed with $(f_2, c_2) = (1.0, 0.9)$
- Take k = 1.

Conclusion (about A — "black things"):

- $f = f_1 =$ **0.9**
- c = (1.0 × 0.9 × 0.8) / (1.0 × 0.9 × 0.8 + 1) = 0.72 / 1.72 ≈ **0.419**
- Power = 0.9 × 0.419 ≈ **0.377**

Compare to what deduction would give with similar premises: Power would be around 0.72. So induction here has roughly *half* the power of deduction. That's the cost of generalizing.

Abduction:

Abduction is a type of reasoning where we move from an **observed result/effect** to the **most likely cause**.

Abduction means finding the best possible explanation for something we observe

Frequency

$$f = f_2$$

Compare this to induction, where $\mathbf{f = f_1}$. The roles have **swapped**.

In abduction, the conclusion's frequency is inherited from the **evidence about B** (the second premise), not from the relationship.

Confidence

$$c = \frac{f_1 c_1 c_2}{f_1 c_1 c_2 + k}$$

The "effective evidence" supporting the abductive conclusion is $\mathbf{f_1 \times c_1 \times c_2}$:

- f_1 — how true the A→B relationship looks
- c_1 — how confident we are in that relationship
- c_2 — how confident we are in our observation of B

If any of these is weak, the conclusion's confidence drops. And the +k in the denominator ensures abduction, like induction, can never reach full confidence that epistemic humility is built into the math.

Power

$$\text{Power} = f \times c = \frac{f_1 f_2 c_1 c_2}{f_1 c_1 c_2 + k}$$

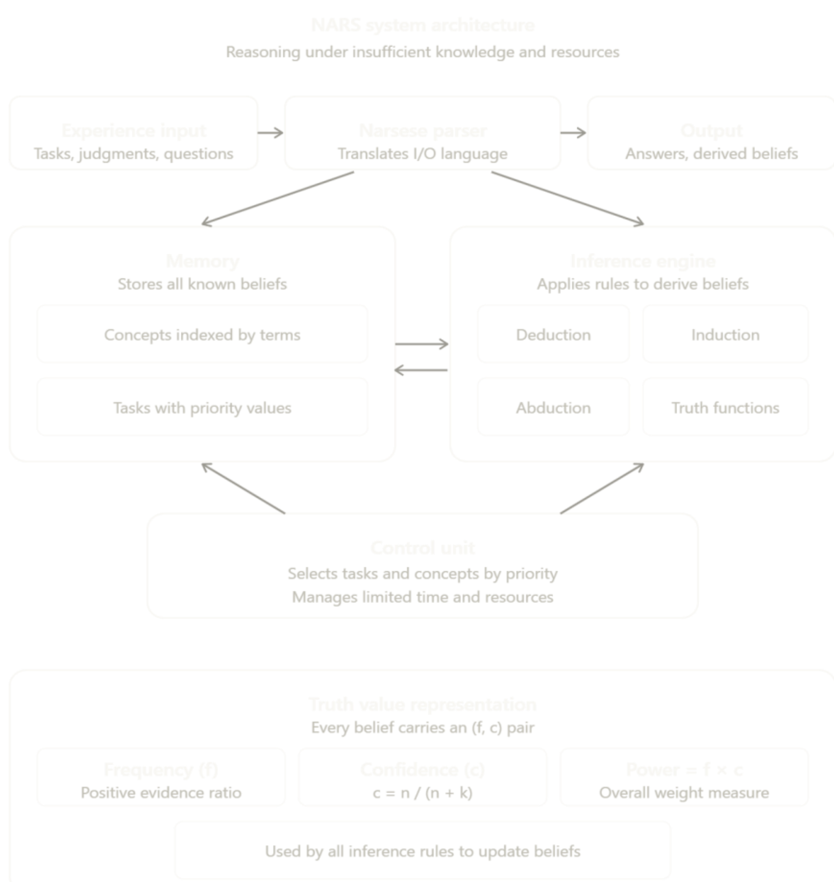
In other words: abduction in NARS is also **non-Bayesian** by design. A Bayesian abduction would require knowing:

- P(rain) — how common rain is
- P(wet grass) — how common wet grass is
- P(wet grass | rain) — how reliably rain produces wet grass

Then Bayes' theorem gives:

$$P(\text{rain} \mid \text{wet grass}) = \frac{P(\text{wet grass} \mid \text{rain}) \cdot P(\text{rain})}{P(\text{wet grass})}$$

But again, NARS refuses to use term probabilities like P(rain) or P(wet grass). So it can't do classical Bayesian abduction. Instead, it uses the formulas above non-Bayesian, but consistent with its philosophy of not assuming what hasn't been observed.



Probabilistic Logic Networks (PLN)

PLN was originally inspired by a desire to capture key aspects of NARS but within a framework compatible with probability theory.

compatible with probability theory Bayesian math, expectations, conditional probabilities, all the standard tools.

The original goal was a kind of **hybrid behavior**:

- When evidence is **scarce** → behave like NARS (evidence-based, non-Bayesian)
- When evidence is **abundant** → behave like classical Bayesian statistics

Scarce means *very little of something* when there isn't enough, or it's hard to find.

Abundant means *a lot of* or *plenty of* something when you have so much of it that there's more than enough.

PLN uses the pair (s, c) instead of NARS's (f, c) and the meanings are subtly but importantly different.

Where:

S strength (s): ***Estimated probability that the statement is true.***
C confidence (c): ***How certain you are about the strength estimate***

This is borrowed from **imprecise probability theory**.

- There's a "first-order" probability the actual probability of the statement being true. PLN's strength s is an *estimate* of this.
- But you're not sure of your estimate. So there's a "second-order" distribution a probability distribution *over what the true probability might be*.
- The mean of that second-order distribution is s.
- Confidence reflects how *peaked* or *spread out* that second-order distribution is.

So before you can even *assign* a term probability, you have to decide: in what universe? In what context? At what scale?
This is a deep problem for PLN
PLN's struggle with term probabilities you can't define probabilities in a vacuum; you need a context.

What this all means for the comparison:

Aspect	NARS	PLN
Truth value pair	(f, c) frequency, confidence	(s, c) strength, confidence
What strength/frequency means	Ratio of positive evidence	Estimated probability of being true
Statement uncertainty	Yes	Yes
Term uncertainty	**No**	**Yes (probabilistically)**
Confidence formula	n / (n+k)	Width of confidence interval over a second-order distribution
Context needed for inference?	Not really	Yes, critically
Philosophical posture	Evidence-counting, no assumptions	Probability estimation, context-dependent

Basic PLN Truth Value Formulas:

"Rather than providing single all-purpose truth value formulas corresponding to inference rules, PLN gives a methodology for deriving truth value formulas corresponding to rules."

NARS	PLN
One fixed formula per rule	Many possible formulas per rule
Derived from axiomatic principles	Derived from explicit assumptions
No tunable choices	Choose assumptions to match your situation

PLN says: "Tell me your assumptions, and I'll give you the corresponding formula." Different assumptions yield different formulas — all of which are valid PLN formulas, just for different scenarios.

I. The **independence-based** formula (the standard one)
II. The **concept geometry-based** formula (an alternative)

Both work with (s, c) truth values, but they're built on different assumptions about how the world works.

The independence-based deduction formula:

The formula

$$\boldsymbol{s_{AC} = s_{AB}s_{BC} + \frac{(1 - s_{AB})(s_C - s_B s_{BC})}{1 - s_B}}$$

This is the correction term. It accounts for the part of A that is not in B but might still be in C through some other path.

Think about it visually:

- Some As are in B (proportion s_AB)
- Those As inherit "C-ness" through B (with probability s_BC)
- But there are also As outside B (proportion 1 − s_AB)
- Some of those might still be in C, just not because of the B connection

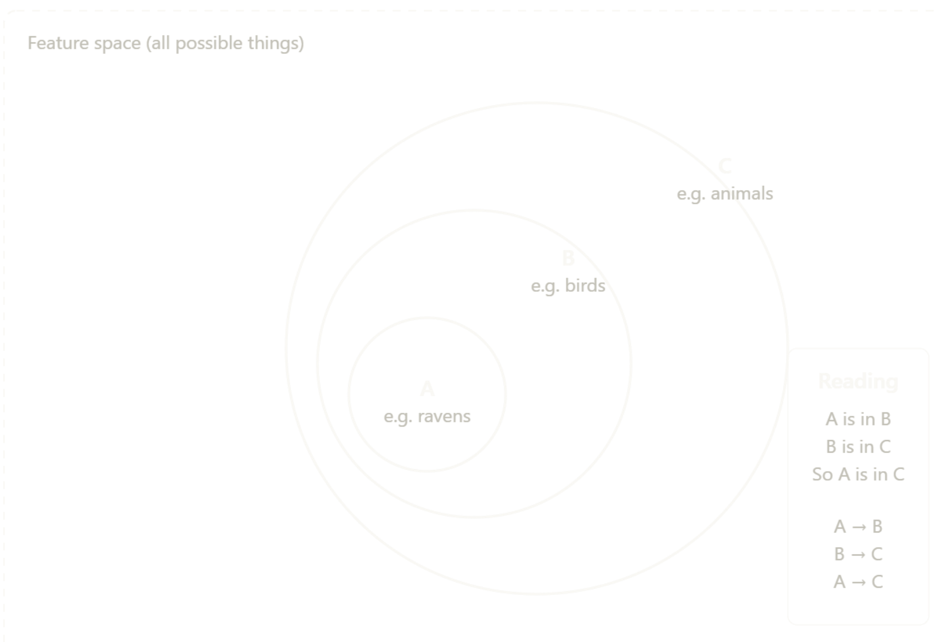
The formula assumes that A and C are probabilistically independent within B's complement that is, "knowing whether something is in A tells you nothing about whether it's in C, beyond what B tells you."

The concept geometry formula

$$s_{AC} = \frac{s_{AB} s_{BC}}{\min(1, s_{AB} + s_{BC})}$$

Imagine a big space of "all possible things." Within that space:

- **A** is a circular region — the set of things that are A
- **B** is a circular region — the set of things that are B
- **C** is a circular region — the set of things that are C



Comparing the two PLN deduction formulas

Aspect	Independence-based	Concept geometry-based
Assumption	Terms are random feature combos	Terms are spheres in feature space
Inputs needed	Premise strengths + 3 term probabilities	Only premise strengths
Complexity	More complex formula	Simpler formula
When it shines	When term probs are well-known	When term probs are unknown or unreliable
When-it struggles	When hidden dependencies exist	When concepts don't actually have spherical structure

Induction:

induction = Bayes + deduction

"The general approach to induction truth value formulas in PLN is to treat induction as a chaining of deduction with Bayes rule."

PLN says: "I already have a deduction formula. Can I reuse it for induction? Yes — if I can first flip the inheritance direction of one of the premises using Bayes' rule."

In other words, induction in PLN is a **two-step process**:

1. **Apply Bayes' rule** to convert "B → A" into "A → B"
2. **Apply deduction** to "A → B" and "B → C" to get "A → C"

Bayes' rule, applied to inheritance relationships in PLN, gives:

$$s_{AB} = \frac{s_{BA} \cdot s_B}{s_A}$$

Why this formula?

Recall that PLN interprets these strengths as conditional probabilities (in a chosen context):

- **s_BA** = P(A | B) — probability that something is an A given that it's a B
- **s_AB** = P(B | A) — probability that something is a B given that it's an A
- **s_A** = P(A) — probability that something is an A
- **s_B** = P(B) — probability that something is a B

Classical Bayes' rule says:

$$P(B \mid A) = \frac{P(A \mid B) \cdot P(B)}{P(A)}$$

Translating to PLN notation:

$$s_{AB} = \frac{s_{BA} \cdot s_B}{s_A}$$

This is literally Bayes' rule with different symbols. That's the whole "flip the arrow" trick.

Step 2: Apply deduction

Now that you've flipped B → A into A → B, you have:

- **A → B** with strength s_AB (from Bayes)
- **B → C** with strength s_BC (the second original premise)

These are the inputs for **deduction**. So you plug them into the deduction formula. Using the independence-based deduction formula:

$$s_{AC} = s_{AB}s_{BC} + \frac{(1 - s_{AB})(s_C - s_B s_{BC})}{1 - s_B}$$

And substituting s_AB = s_BA s_B / s_A:

$$s_{AC} = \frac{s_{BA}s_{BC}s_B}{s_A} + \frac{(1 - s_{BA}s_B/s_A)(s_C - s_B s_{BC})}{1 - s_B}$$

After algebraic simplification, the authors get:

$$s_{AC} = \frac{s_{BA}s_{BC}s_B}{s_A} + \frac{s_B(1 - s_{BA})(s_C - s_B s_{BC})}{s_A(1 - s_B)}$$

This is the **full PLN independence-based induction formula**

NARS vs PLN

Aspect	NARS	PLN
Developer	Pei Wang	Ben Goertzel and colleagues
Philosophical basis	Non-axiomatic, evidence-based reasoning	Probabilistic reasoning grounded in probability theory
Truth value pair	(f, c) = frequency, confidence	(s, c) = strength, confidence
What strength/frequency means	Ratio of positive evidence to total evidence	Estimated probability that the statement is true
What confidence means	c = n / (n + k), based on evidence count	Width/narrowness of confidence around a second-order probability distribution
Can confidence reach 1?	No, always less than 1	No, always less than 1
Uncertainty on relationships?	Yes	Yes
Uncertainty on terms?	No	Yes, term probabilities
Term probability inputs needed?	None	Needs term probabilities such as s_A, s_B, s_C
Requires inference context?	Not strictly	Yes, critically
Foundation for inference	Direct evidence-based formulas	Probability theory + Bayes' rule
Independence assumption?	Avoided explicitly	Used in standard deduction formula

Main Fundamental Difference

System	Where uncertainty is attached
NARS	Only to statements / relationships
PLN	To both terms / concepts and relationship